

Phase 4 - Logging, Journald Persistence & Time Sync

This document covers centralized logging configuration (rsyslog and journald), a diagnosed facility-routing discrepancy, the SSH login evidence tying Phase 3 to Phase 4, and time synchronization via chrony.

1. rsyslog Facility Routing Investigation

An initial test using `logger -p local7.notice` did not appear in `/var/log/messages` as expected. Investigating `/etc/rsyslog.conf` revealed the cause:

```
local7.* action(type="omfile" file="/var/log/boot.log")
```

Finding: `local7.*` is explicitly routed to `/var/log/boot.log`, RHEL's convention for boot-related messages, rather than the general catch-all file. The test message was confirmed present in the correct file, validating that rsyslog's facility-based routing was working exactly as configured — the initial assumption about default routing, not the logging pipeline itself, was incorrect.

2. journald Persistence

journald was switched from its default volatile (RAM-backed) storage to persistent on-disk storage by setting the following in `/etc/systemd/journald.conf`:

```
[Journal]
Storage=persistent
```

After restarting `systemd-journald` and running `journalctl --flush` to migrate the in-memory runtime journal to disk, the persistent journal directory was confirmed present, named after the system's machine ID:

```
4cbd8cbcd7fb443f991ab90abaf25410
```

Note: A plain service restart after the config change was not sufficient on its own — the runtime journal required an explicit `journalctl --flush` to actually relocate from `/run/log/journal` (volatile) to `/var/log/journal` (persistent).

3. SSH Login Evidence — Phase 3 + Phase 4 Tie-In

With persistent journald storage confirmed, the full SSH authentication history from Phase 3 was retrieved via `journalctl -u sshd`. This log captures the complete story: an initial failed password attempt, the expired-password enforcement from Phase 2's aging policy, a password-based login, and finally the key-based login established in Phase 3:

```
Jul 04 18:16:22 servera.mylab.com unix_chkpwd[6930]: password check failed for user (alaa)
Jul 04 18:16:22 servera.mylab.com sshd-session[6928]: pam_unix(sshd:auth): authentication failure;
logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.136.128 user=alaa
```

```

Jul 04 18:16:24 servera.mylab.com sshd-session[6928]: Failed password for alaa from
192.168.136.128
port 60166 ssh2
Jul 04 18:16:30 servera.mylab.com sshd-session[6928]: pam_unix(sshd:account): expired
password for
user alaa (root enforced)
Jul 04 18:16:30 servera.mylab.com sshd-session[6928]: Accepted password for alaa from
192.168.136.128 port 60166 ssh2
Jul 04 18:19:43 servera.mylab.com sshd-session[7119]: Accepted password for alaa from
192.168.136.128 port 57962 ssh2
Jul 04 18:19:44 servera.mylab.com sshd-session[7119]: pam_unix(sshd:session): session
opened for
user alaa(uid=1005) by alaa(uid=0)

```

The key-based authentication event — the definitive proof that SSH key login (Phase 3) is durably preserved through journald's persistent storage (Phase 4) — is highlighted below:

```

Jul 04 18:22:07 servera.mylab.com sshd-session[7195]: Accepted publickey for alaa
from
192.168.136.128 port 36868 ssh2: ED25519
SHA256:o0nw5S52V8fWl4JaDD0FTqUMFZRRZeL5ap13isTIVyk

```

```

lobnaahmed@servera:~$ ls /var/log/journal/
4cbd8cbcd7fb443f991ab90abaf25410
lobnaahmed@servera:~$ ls /var/log/journal/4cbd8cbcd7fb443f991ab90abaf25410/
system.journal user-1000.journal
lobnaahmed@servera:~$ cat /etc/machine-id
4cbd8cbcd7fb443f991ab90abaf25410
lobnaahmed@servera:~$ journalctl -u sshd -n 20
Jul 04 18:16:22 servera.mylab.com unix_chkpwd[6930]: password check failed for user (alaa)
Jul 04 18:16:22 servera.mylab.com sshd-session[6928]: pam_unix(sshd:auth): authentication failure;
Jul 04 18:16:24 servera.mylab.com sshd-session[6928]: Failed password for alaa from 192.168.136.128
Jul 04 18:16:30 servera.mylab.com sshd-session[6928]: pam_unix(sshd:account): expired password for
Jul 04 18:16:30 servera.mylab.com sshd-session[6928]: Accepted password for alaa from 192.168.136.1
Jul 04 18:19:33 servera.mylab.com sshd-session[7098]: Connection closed by authenticating user alaa
Jul 04 18:19:33 servera.mylab.com sshd-session[7105]: Connection closed by authenticating user alaa

```

4. Time Synchronization (chrony)

System time synchronization was enabled and verified. timedatectl confirms the clock is synchronized and NTP is active:

```

Local time: Sat 2026-07-04 20:43:15 EEST
Universal time: Sat 2026-07-04 17:43:15 UTC
RTC time: Sat 2026-07-04 17:43:15
Time zone: Africa/Cairo (EEST, +0300)
System clock synchronized: yes
NTP service: active
RTC in local TZ: no

```

4.1 chronyd Service State

- `chronyd.service` - NTP client/server
 - Loaded: loaded (/usr/lib/systemd/system/chronyd.service; enabled; preset: enabled)
 - Active: active (running) since Sat 2026-07-04 20:36:21 EEST; 6min ago
 - Main PID: 8480 (chronyd)

```

Jul 04 20:36:21 servera.rhelproject.local chronyd[8480]: Frequency -50.822 +/- 0.055
ppm read
    from /var/lib/chrony/drift
Jul 04 20:36:26 servera.rhelproject.local chronyd[8480]: Selected source
176.97.192.150
    (2.rhel.pool.ntp.org)
Jul 04 20:36:26 servera.rhelproject.local chronyd[8480]: System clock wrong by
1.752673 seconds
Jul 04 20:36:28 servera.rhelproject.local chronyd[8480]: System clock was stepped by
1.752673
seconds

```

4.2 Active NTP Sources

MS Name/IP address	Stratum	Poll	Reach	LastRx	Last sample
=====					
^* 139.84.231.68.vultruserc>	2	6	377	18	+145us[-227us] +/- 111ms
^+ reserved.vm	2	6	377	17	+313us[+313us] +/- 221ms
^+ mirror.xcobebean.co.ke	2	6	377	15	-2555us[-2555us] +/- 168ms
^- ntp3.inx.net.za	2	6	377	16	-2294us[-2294us] +/- 145ms

The '*' marker identifies the currently selected best time source (139.84.231.68, a public NTP pool server), with three additional sources providing cross-verification.